

# **To what extent can color psychology influence consumer perceptions of product quality?**

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## **ABSTRACT**

Actually color does play a significant role in consumer decision making, as it has the ability to confuse consumers before they even touch or use a product. The present paper examines the extent to which color psychology influences consumers' perceptions of product quality from a combination of reviewed theories, empirical studies in the real world, and a few examples of industry type. Rather than using colour as decoration, the paper frames colour with concepts such as: Signaling Theory, Cue Utilization Theory, and Processing Fluency Theory, which all ask the question of how colour can be used to guide emotional responses, cognitive evaluation and quality assumptions. The literature indicates that colors may carry certain meanings and feelings such as trust, luxury, nature, reliability, etc., which are relied on when there is not much information. You can see this in evidence from packaging design, luxury branding, food and beverage marketing , and also in online shopping situations. In those regions, it appears color can influence what people believe of the quality, the trustworthiness of a product and the value feeling “right.” But, even more importantly, the illustrations of Apple, Tiffany & Co., Whole Foods, Coca-Cola, and Pepsi show that the industry is aware of and uses color purposefully. Colors can not only be used to place the brand, but they also can suggest characteristics connected to quality – but not necessarily explicitly. However, the results are not entirely consistent and clear-cut. The impact that the colour will have is likely to differ based on the individual, the type of product and any

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branding cues already given such as pricing. Colour psychology can have a significant influence on the appraisal of quality of a product, but this is dependent on the situation.

**Keywords:** color psychology, consumer perception, perceived quality, branding, packaging, consumer behavior, visual marketing

## 1. INTRODUCTION

The whole growing importance of visual communication in modern markets has kinda reshaped how consumers judge products and brands, it's not just a small change. Every aspect of purchasing is becoming competitive and increasingly digital, meaning that people are often forced to make decisions with only a handful of pieces of information. And for many people, particularly online retailers, customers cannot truly judge things by the look of them before purchase, and rely on visual cues that are provided via the product images, packaging, brand style and even the structure of the website. Due to this, these visual elements have become an integral component in marketing plans designed to affect consumer perceptions

and purchasing decisions (Underwood, 2003). Of all those visual elements, color has become one of the most powerful ones that impact the perception and appreciation of products.

Then digital commerce enters the picture and boosts it to an entirely new level. Over the past ten years, global e-commerce sales have significantly increased, making brands increasingly turn to digital product displays, as well as to visual branding, to communicate to their customers about the appearance of the product and the value it possesses. Unlike traditional brick-and-mortar shoppers, online users aren't able to physically feel, test or thoroughly examine a product before buying. Thus, product packaging, website appearance, and other

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visual design elements become readily available sources of information that allow consumers to make judgments about product quality and also about reliability (Bleier et al., 2019). In these types of settings, color becomes a readily available signal, and can swiftly guide perceptions and expectations.

The findings from consumer psychology research indicate that consumers are able to make impressions within seconds only after seeing something visual. The majority of these fast calls rely on heuristic processing, or simply a mental shortcut where people rely on their gut instincts so they can make a quick decision, particularly when information is not available in full (Kahneman, 2011). Visual cognition aids consumers to quickly understand and evaluate products, often before they enter into thoughtful analysis of features. Throughout all this, colour appears to be a significant factor as it is one of the first things that is noticed and can evoke emotional and cognitive

reactions almost immediately (Labrecque & Milne, 2013).

But colour influence extends beyond 'looks'. Consumers frequently rely on color to make judgements about a product's reliability, luxury, safety, value and even overall quality. For example, blue is often perceived as trustworthy, competent, black as sophisticated and prestigious. Green is often associated with 'green' or 'heathy' while red could represent excitement, energy, or urgency (Elliot & Maier, 2014). These links can steer what consumers expect from a product, before they've had any real firsthand experience with performance. As a result, marketers will often select colours deliberately, to align with the positioning of their brands and influence the impression of quality.

While color psychology is quite commonly applied in marketing and branding, it seems like the relationship between color and consumer perception is still a bit confusing. Colors do not have constant meanings, do not have universal meaning,

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and even change meaning within a culture, among various demographic groups and even among various contexts in which they are consumed. A colour that appears high quality or prestigious in one country, may represent something very different in another. Moreover, the effect of one color is not always the same as different colors may be suitable to different categories of products being assessed. For instance, black packaging may give a luxurious appearance to cosmetics or electronics, but not necessarily so with food products. As such, properties such as brand reputation, product type, and expectations of the local culture can influence interpretation of the color, and whether it actually leads to a quality judgment (Aslam, 2006).

Given all that, it's still important to take the time to examine how much really lies in the realm of color psychology to affect what consumers consider “good” about a product. Many companies invest in colour-based branding and packaging strategies, but there are many questions regarding the

durability of such effects, and the impact of such marketing in different environments and amongst different consumer groups.

Thus, the present paper aims to provide an answer to the following research question:

**To what extent can color psychology influence consumer perceptions of product quality?**

The study aims at answering the question in four steps. First, it examines large-scale theories that attempt to account for the links between color and consumer perceptions. Second, it validates academic research on the relationship of color to peoples' evaluation of product quality. Third, it delves into industry examples of when and how organizations sort by color in their branding and packaging. Last but not least, it assesses the limits of colour psychology, taking into account context and cultural factors that may moderate or affect the impact of colour, according to the context.

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For several reasons this topic is important. Consumers' color perception can also be used to help with marketing strategies, such as developing effective branding and communication campaigns. Knowing how colour affects people can help in designing packaging that uses colours as a guide to the positioning of the product. It's also consistent with studies on consumer behavior, which show that when there is a limited amount of information, people will rely on outside cues to make their decisions. This paper will build upon the landscape of effect of visual cues on product perceived quality in the modern market, exploring what colour psychology can achieve, and what it cannot.

## **2. CONCEPTUAL FOUNDATIONS AND THEORETICAL FRAMEWORK**

To grasp the impact of colors on consumers' quality perceptions requires some kind of scaffolding of theory to describe how the visual signals are converted into decisions. Color isn't just random, it is more like the effect can be

seen cognitively, emotionally and perceptually, assisting consumers in making sense of what they see. In this part, i introduce the main ideas and theories that will be the bedrock for looking at how color psychology relates to perceptions of product quality.

### **2.1 Understanding Color Psychology**

Color Psychology is the study of the effects colors have on the human experience, including feelings, thought patterns, and behaviors. Color psychology is the study of the influence of different colors on consumer perceptions, attitudes and purchasing decisions (Elliot & Maier, 2014). The concept is based on the premise that colours have learned meanings and past experiences which can influence psychological responses, often subconsciously.

**Figure: Major color associations from Literature**

| Color                                                                                   | Common Association |
|-----------------------------------------------------------------------------------------|--------------------|
|  Blue  | Trust              |
|  Black | Luxury             |
|  Green | Natural            |
|  Red   | Excitement         |
|  White | Purity             |

*Source: Author*

Colour perception is closely related to this, and refers to how people perceive, interpret and meaning assign visual colour stimuli. The process of colour perception starts with physiological responses in the visual system but the final interpretation is influenced by factors such as personal experience, culture and context of presentation (Labrecque et al., 2013). Therefore, consumers may assign different meanings to the same colour, depending on the context, the place and the moment the colour appears.

Colour is, like, tightly bound up with emotional feelings. Some colors seem to

have a pretty consistent effect on bringing certain emotions up. For example, blue is associated with trust, security, competence and green with nature, health and sustainability. While Black is typically associated with luxury and refinement, red is often seen as vibrant, spirited and urgent (Elliot & Maier, 2014). And these emotional links can end up steering how consumers judge a product in advance of any real, hands on experience.

There are three important angles that are typically relevant to how people experience colour: hue, saturation and brightness. Hue is the color family, such as red, blue, green or yellow. The intensity or the purity of the color is called saturation - a colour with high saturation is bright and strong, while lower saturation is subdued. Brightness refers to the overall lightness or darkness of a colour. But when these components change, so do the consumer responses. Darker hues are generally interpreted as expensive and prestigious, while lighter hues are associated with

welcoming, economical or even energetic (Labrecque & Milne, 2012). In other words, these attributes contribute to how people perceive products and brands.

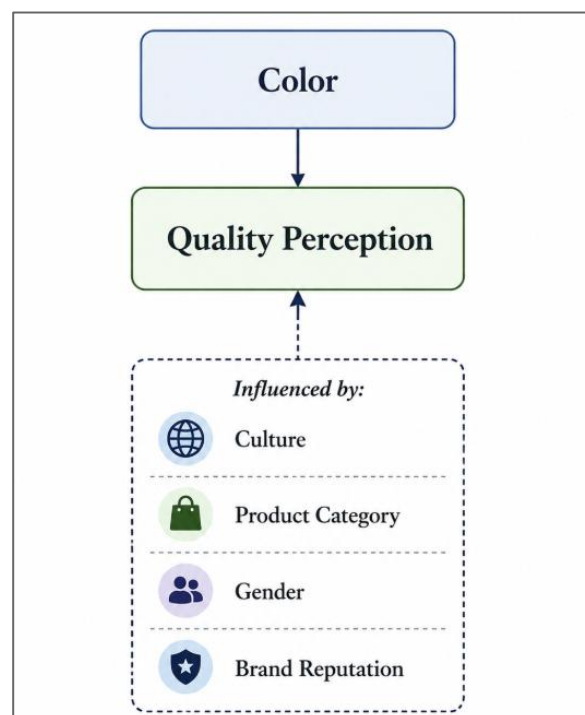
The effect of color on the mind occurs because colour is a symbolic suggestion that arouses pre-existing associations and emotional responses. Consumers learn what certain colors are “supposed” to be over time, as they see and hear these colors in culture, in marketing and in daily life. This makes colors powerful tools to set perceptions, even prior to a consumer's conscious thought of product details.

## 2.2 Consumer Perception and Product Quality

To understand how color affects judgments of quality, it helps to sort out objective quality versus perceived quality a bit. Objective quality is basically the real, testable sort of performance or features of a product like durability, reliability, ingredients, or technical specs (Zeithaml, 1988). These can typically be

ascertained by testing or by hands-on experience with the product.

**Figure: Moderating Factors Influencing the relationship between Color and Quality Perception**



*Source: Author*

Perceived quality, on the other hand, is the customer's personal judgment on the quality of the product, whether it be good or simply better than others. Of course, this could be out of step with what the product can actually do. The definition of quality is formed by the individual prior to

use, and is related to what is available at the moment of purchase (Zeithaml, 1988).

This distinction is significant as buyers rarely have complete information. Typically, they do not sit down and consider all the attributes. Instead they take cues from outside that assist them in estimating quality in a timely manner. Those cues can be cost, brand reputation, packaging design, trust of the retailer and even just the appearance. That is, they serve as a sort of signpost in place of quality, particularly if the customer has some uncertainty.

Colour is one of the first things people notice, so it fits in nicely. A dark, sophisticated color on a product may give it a high-end look. However, if the same type of product is bright, fun colour, it could mean it is of lower quality, or possibly targeting younger customers. And these impressions can occur even if the objective information remains unchanged.

## **2.3 Signaling Theory**

The Signaling Theory is one of the most useful theoretical perspectives to understand the influence of color on quality perceptions. It began in economics and it kind of illustrates how people make decisions when they don't have all the information or if it's not being disseminated evenly across the sides (Spence, 1973). In consumer markets, as in most markets, there is typically much more knowledge regarding product quality available to the company than available to the consumer; the consumer then resorts to using “signals” to direct the choice process and alleviate the uncertainty.

Typical quality signals include price , brand reputation, warranties, packaging design, and marketing communications. They did this together, allowing consumers to make a judgment about quality without having to test the product first. Research indicates that people tend to rely on visual cues because they're there, they're easy to

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find, and they're not that hard to interpret (Kirmani & Rao, 2000).

In this concept, color can also be used as a quality indicator. Some colors are essentially bound to certain qualities of the product over time and through a sort of cultural learning. For example, black and gold are often associated with high quality items whereas white is often associated with cleanliness, purity and safety. When consumers encounter these colors, they might assume similar qualities, even before examining the details. For this reason, colour helps communicate quality by providing quick visual cues that inform expectations, and then subsequent evaluations.

## **2.4 Cue Utilization Theory**

Another, sort of influential theory is Cue Utilization Theory - how people use cues they can obtain to make a judgement about a product. That is, they are able to explore the information cues that are available, to assess what they see and what they may be

purchasing. According to that theory, when people are trying to make a choice or judgment about quality they rely on both intrinsic and extrinsic cues (Olson & Jacoby, 1972).

Intrinsic cues are simply the characteristics of the product itself, and they go hand-in-hand with the product's performance. Materials, ingredients, taste, durability, technical functionality, for example. These signals typically provide the best insight into the actual quality of the product. However, they may not be easily identifiable or available prior to making purchases.

This is the reverse, external aspects that aren't a natural component of the product itself, but that still influence consumer assessments. Consider packaging, labels, branding, advertising, country of origin and even color. Extrinsic cues are often the only information that consumers have at the time of sale, making them more likely to use them to make a guess about the intrinsic information.

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So, colour becomes an important extrinsic cue, primarily because it sets expectations before any use of the product. Green packaging can have a positive connotation as a suggestion of natural ingredients or sustainability, and darker colours can be interpreted as an indication of luxury or quality. These small associations can have a strong influence on quality perception where product attributes are not known.

## **2.5 Processing Fluency Theory**

Processing Fluency Theory is a newer explanation for the effects of color on consumer perceptions. Basically, this suggests that human beings are more receptive to information that is simple to understand and ponder (Reber et al., 2004). Spotlights that are familiar or expected help shoppers experience more cognitive ease, which often steers them in a better direction in evaluating.

The colours in marketing that match the consumers' existing perceptions of that product line are generally processed

quicker than unexpected combinations. Therefore, any object that looks like it should could appear more real, more appropriate, and even better. For example, dark colour packaging is associated with luxury chocolates, primarily because the dark colour packaging has become a part of the standard in the luxury confectionery market. Similarly, white and blue packaging is visible on a large scale for healthcare and hygiene products as the colours are associated with cleanliness, safety and professionalism.

If the colors align with consumers' expectations, it makes the product look more familiar, and easier to understand. This “easy processing” good feeling can then transform into an improved “quality” feeling. Colors that are not consistent with those of the category, however, might feel out of place, leading to confusion and lower positive evaluations. The key concept of processing fluency theory is that colors can have a profound effect on

consumer choices, even if the consumer isn't directly interacting with the product.

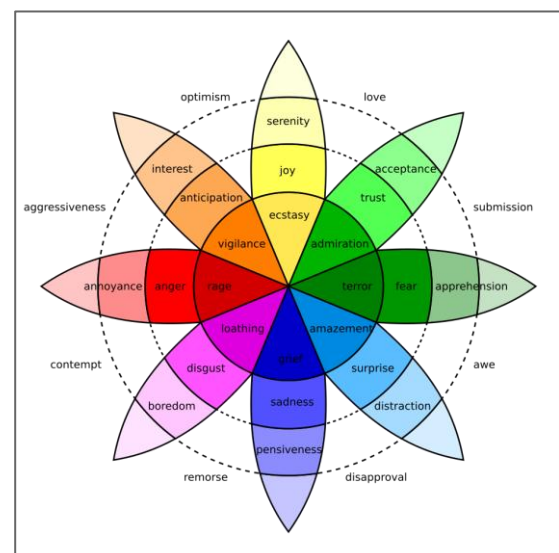
### 3.MECHANISMS THROUGH WHICH COLOR INFLUENCES QUALITY PERCEPTIONS

The relationship between colour and people's impression of the quality of a product is no accident and it is not by chance. However, it does so in several psychological ways, and that is how consumers interpret and then judge what they see. In reality, color can evoke feelings, trigger mental shortcuts, reinforce the consumer's expectations from the product category and can make brand recognition seem quicker. These parts tend to be used concurrently, so the customer decides before actually having a chance to see or experience the product's true capabilities. These processes are important to understand because they help to explain why colour has become a powerful marketing, branding and packaging tool.

### 3.1 The emotional responses to color.

One of the most researched methods of colour influencing consumer perceptions is to evoke an emotional response. Psychology and marketing research studies have shown that colours can have symbolic meanings. These meanings are created through both biology and social learning processes (Elliot & Maier, 2014). These meanings can be activated to trigger emotional responses which will then influence the consumer's product and brand perceptions.

**Figure: Plutchik's Wheel of Emotions**



Looking at the most common colors in marketing, blue emerges repeatedly as the

color of trust, reliability, competence and professionalism. As a result of those connections, many financial institutions, technological corporations, healthcare providers and larger companies rely on blue for stability and reliability. Labrecque and Milne (2012) found that the consumer often associates the blue brand with competence and trustworthiness, and these attributes have a positive impact on the perceived product quality. If the consumer thinks a brand is reliable, then he or she has a better idea that its products are well engineered and will perform consistently over time.

Black is a color that definitely has a strong association with people's perception of quality. For consumers, black is a symbol of sophistication, prestige, exclusivity and luxury. Many luxury brands have black packaging and branding as it is an indication of luxury and elegance. Research has indicated that darker shades create an impression of higher value and quality, particularly in industries such as

cosmetics, fashion, cars and electronics (Madden et al., 2000). Even if the product remains the same, consumers can begin to believe that the product in black packaging is better.

Green has also grown increasingly relevant in the current markets owing primarily to the growing environmental consciousness and the interest of consumers in their well-being. Green is typically associated with nature, freshness, organic farming and the environment. These meanings can influence product quality perceptions by signaling the product's natural ingredients, responsible or ethical approaches to product manufacturing, and health benefits (Aslam, 2006). In the food and beverage industry, for example, green packaging can create the impression that it is healthier and more environmentally friendly than other food and beverages with the same color patterns.

Red elicits other emotional reactions. It's often associated with excitement, passion, power, energy and urgency... well it does

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attract attention! According to Elliot and Maier (2014), red is a color that has been found to have a more effective impact on the arousal of a group of people and get their attention more than other colors. The effect of red on perceived “goodness” is related to other factors besides the color itself, such as the context in which it is used, even if the outcome is perceived as dynamism and strength. For certain product categories, red might be a sign of confidence and more expensive performance, while in other cases it may be perceived as impulsiveness or deal-hunting, more of a sales feature than a true quality.

These examples also show that people seldom react to color purely as a visual thing. Rather, colors turn on emotional associations that then steer expectations about products. If the consumers experience positive emotions like trust, prestige, or confidence in the process, they are likely to transfer these positive emotions to the perception of product

quality. As a result, emotions are a crucial pathway by which color influences consumer perceptions, even if the consumer is not directly aware of it.

### **3.2 Cognition Heuristics and Mental Shortcuts**

The second is based on cognitive heuristics – mental shortcuts that facilitate consumers' decision making in situations where information is scarce. The modern consumer is today confronted with massive amounts of product information and it is difficult to conduct a detailed evaluation of each of them. Because of that, they frequently fall back on simple cues, that basically make the decision process faster (Kahneman, 2011).

Color is like one of those cues, as it can be processed nearly instantaneously. As early as several seconds after exposure, some studies indicate that consumers begin to make an impression and that they often rely on the visual characteristics before even considering technical specifications

or reading through detailed product descriptions (Orth & Malkewitz, 2008). Such quick procedures are typically utilized in online shopping scenarios, where individuals could possibly explore hundreds of products in a matter of moments, similar to a blink.

Automatic thinking is also closely linked to the use of colour as a heuristic. Consumers' decisions on whether a product is attractive and trustworthy come from intuitive judgments rather than thoughtfully judging why it looks good or is trustworthy. For instance, a healthcare item packed in white and blue shades might be easily thought of safe and professional, since white and blue are frequently linked to cleanliness and medical knowledge. Likewise, a luxury product in the black pack will be perceived as a premium product until its clients know about its premium features and premium price.

Research into packaging design has revealed a bit about how visuals can

establish a good expectation of what the packaging does, and how it feels. Orth and Malkewitz (2008) state that packaging components do impact on the consumer perception – and sometimes before the product is even used. Setting these expectations can impact the evaluation of the product in the future. For example, consumers may be more likely to interpret the actual experience as positive, if they think that the colour is an indicator of high quality experience, highlighting the role of heuristic processing in consumer decision making.

This kind of reliance on visual cues is significant because there are many transactions made when people are in a state of uncertainty. Typically, it is impossible for individuals to be able to verify the quality of a product, especially prior to purchase. So, color becomes a convenient cue—or just a simple hint—that helps to minimize uncertainty, and then guides assessments down the proper path.

### 3.3 Color-Category Congruence

A finding that has been found to be strong and statistically significant in color psychology research is what is known as color-category congruence. In this context, it refers to the extent to which a product's colour match is in line with what consumers presume for that type of product.

Consumers develop their own notion of colors “fitting” certain products, typically because they continue to encounter the same patterns over and over again across the industry and via general market conventions. For example, organic food products tend to be green, as it sort of indicates nature, as well as sustainability. Black is associated with luxury products and it seems like exclusivity and more elegant and sophisticated look. White and blue colors are used frequently in healthcare products, as they evoke cleanliness and safety, and an air of professionalism. As far as kids stuff, bright, highly saturated colors are used

quite often as those colors are read as fun, energetic and playful.

Other studies suggest that when the product's colour is similar to what people “expect” from that type of product, assessments tend to be more favourable. Then there's Processing Fluency Theory, which states that colours that match the type of product make it easier to process, because the colours are associated with the mental images people have of the product type (Reber et al., 2004). That it might be easier processing can lead to a sense of familiarity, and the selection seems more appropriate could drive up perceptions of quality.

However, when the colors do not match the typical one of the category, consumers may become confused, if not uncertain. The neon-pink colour scheme may make it appear less luxurious than it would in a top quality perfume, and that's a luxury product. Same concept, with healthcare, if the presentation is towards dark, aggressive colors, it is more tend to look

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untrustworthy than a presentation with white, blue packaging.

The effectiveness of colors, which is dependent on their meanings in relation to the product and consumer expectation, was demonstrated by Bottomley and Doyle (2006). Put differently, they didn't conclude that the successful color strategy is to select something that looks attractive. It's more about picking out colors that also highlight the product positioning you're trying to achieve, so to speak, like fitting the message in, you know. Thus, the pathway of color–category congruence emerges as a significant route, by which color is seen to influence quality judgements.

### **3.4 Brand Identity and Quality Inference**

Colour is crucial not only for the one product, but also as an element of the quality perception in the process of branding. When using colours with a consistent application for all brands, it is

possible to create a more differentiating visual identity that can be easily recognised and remembered. If the same brand images are present for a long period of time, the familiarity is conducive to establishing trust, which can further reinforce this trust.

Brand recognition is also important as well-known brands are perceived as less risky than unknown brands. It's familiarity that translates to credibility, stability and market success in the eyes of consumers (Keller, 2013). If a brand consistently returns to the same selection of colors in its products, advertisements and packaging, then the colors will be associated with the brand's reputation, and what the brand is reputed to be.

Moreover, color and kind of are a factor in trust development. Having uniform presentation of material has been found to increase the appearance of professionalism and reliability, which have been associated with perceptions of quality (Labrecque et al., 2013). If consumers can keep getting



the same colors and having a positive experience, then their faith in the product's performance starts to grow—and more than it did previously.

Further, color is a reminder to memory – which helps maintain meaning of brands. With time, some of the color traits may be perceived as some of the qualities. Take a fancy brand, for example, a luxury brand. If it is always leaning towards black and gold, then those colours could be a large message of exclusivity and workmanship. When these are established, they can influence subsequent ones even if the individual doesn't actually have very much new information about the new products that the brand launches.

That is why color and perceived quality aren't just based on a “first impression” or even a mental “short-cut.” The colour over time starts to become a part of the brand, it starts to effect perceptions of trustworthiness, familiarity and quality more broadly. Ultimately, colour becomes part of the consumer's buying decisions

both at product and brand level. This is why colors are considered to be a valuable part in the modern marketing strategy and not just as decoration.

Here, color is presented as having a number of interrelated ways to effect quality views in all of the mechanisms presented here. These are all emotional associations, all of which equal a chip in: heuristic processing, category fit, and brand identity. They can be employed to explain how colour is one of the most influential visual reminders in consumers' minds to consider as they choose quality products.

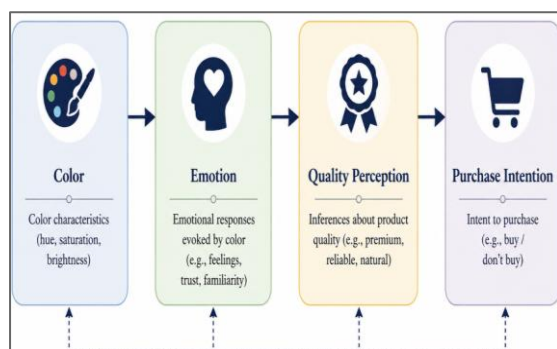
#### **4. EVIDENCE FROM ACADEMIC STUDIES**

While theoretical constructs may help to explain why color may change perceptions of product quality, there is a need for empirical research to determine whether or not those effects actually manifest in decision making. Throughout the last few

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decades, there have been numerous studies conducted by researchers from the fields of marketing, psychology and consumer behavior, examining the relationship between color and product evaluation. Although the results vary in some cases depending on context, there are good lines of evidence suggesting that colour may be used to strongly influence opinion of the quality, trustworthiness, prestige and the perceived performance of a product. This section examines some of the important academic research and categorizes knowledge into four broad themes: packaging design, luxury perceptions, food and beverage evaluations, and online shopping environments.

**Figure: Role of COLOR in shaping purchase intention**



*Source: Author*

#### 4.1 Color and Packaging Studies

Color is one of the most influential settings that can impact consumer thinking, particularly since packaging is often the first product touchpoint consumers encounter. In stores or other retail locations, the colour of the packaging often acts like a kind of tip-off, and may make an initial impression before readers fully engage with the details.

Orth and Malkewitz (2008) present some pretty solid evidence about the connection between package design and consumers' first impressions. In a few empirical studies, they say that if the entire package seems “holistic” in its message, it can make a brand personality difference that is noticeable in the minds of the people. Thus, when packaging is done using natural visual elements, it tends to come across as sincere, whereas a more refined and subtle packaging design is typically associated with competence and a certain sophisticated touch. Although they considered package design as a whole,

rather than just color, color remains a part of the picture, and it is certainly playing a role in package evaluations. In general, their findings indicate that these visual elements of packaging truly influence the perception of product quality and brand image before the product has been consumed.

Similarly, Bottomley and Doyle (2006) investigated the relationship between colour and product category and its impact on brand perceptions. In their experiments, they found that consumers tend to rate the brands more positively when the colors correspond to their brand expectations of the product. That is, if the colors correspond to what people think a category should be, those colors are more appropriate and effective than colors that are inconsistent with people's expectations. All these results support the hypothesis of color as a quality signal, rather than as decoration or as something of pure aesthetic value.

Together, these studies suggest that color is a factor in quality judgements which sends symbolic messages and confirms expectations of product attributes.

### Academic Studies Examining Color and Product Evaluation

| Study                              | Focus                                | Key Findings                                                                                                               |
|------------------------------------|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Orth & Malkewitz (2008)            | Package design and brand impressions | Visual package characteristics influenced perceptions of competence, sophistication, sincerity, and overall brand quality. |
| Bottomley & Doyle (2006)           | Color-product congruence             | Colors consistent with product positioning were evaluated as more appropriate and favorable.                               |
| Labrecque & Milne (2012)           | Brand color meanings                 | Blue was associated with competence and trust, while red generated perceptions linked to excitement and stimulation.       |
| Labrecque, Patrick, & Milne (2013) | Review of color research             | Color influences consumer judgments through symbolic associations, emotions, and learned meanings.                         |

*Source: Author*

### 4.2 Luxury Perception Studies

There is a lot of research out there that indicates that certain colors are truly effective at communicating luxury, prestige and exclusivity – sort of on their own. Of those, dark colors such as black, gold and others are the ones that are mentioned with the greatest frequency when people peak of the highest quality.

The meaning of colors continues to be studied and the same concept keeps repeating itself: black is associated with sophistication, power, elegance and

exclusiveness (Madden et al., 2000). It explains why brands in industries such as beauty, clothes, cars, and tech, are so often enveloped in black packaging and branding. Even before they know any more, many consumers see darker shades as indicators of higher quality and worth.

Labrecque and Milne (2012) also found that brand personality perceptions can be influenced by color and that it can impact consumer evaluations of a product. When colors are congruent with such traits as prestige or competence, individuals are more likely to perceive the item as being high quality. Black and gold are popular colors in luxury areas to convey those attributes. Historically, gold has been associated with power, prosperity, and exclusivity, which is why it's often a symbol of wealth.

Further studies in visual branding have shown that luxury impressions intensify if multiple visual cues are used in combination, not individually. When a product is shown in black packaging with

minimalist design elements, plus a premium price point it's more likely to be viewed as high quality than the identical product styled with bright or highly saturated colors. This is consistent with Signaling Theory discussed above as these visible signals are being used by consumers to derive quality information when it is not available or not readily apparent.

The evidence is therefore that colour influences luxury perceptions, but not because the colour itself is meaningful or because of anything like that, but because people have learnt to associate certain colours with “prestige” products and high status consumption. To some degree it is indirect, but yes, that's the point.

#### **4.3 Food and Beverage Studies**

Some of the best examples of how color affects consumer perceptions or expectations can be seen in the food and beverage industry. Numerous studies demonstrate that colour influences

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expectations of taste, freshness, quality and even of the way things "should" work for a product.

Spence (2015) has provided evidence for the influence of visual information on directing sensory expectations. People sometimes pick a product, make a decision regarding flavor and freshness, and then decide on the quality of the product. In most instances, those estimates are influenced substantially by what they see, such as the visual appearance as well as the packaging design. Often colors are signals, they allude to the nutritional value of ingredients and freshness.

The use of green packaging comes up a lot and it's a particularly good example. It has been found that consumers often associate green with naturalness, environmental protection and healthfulness (Aslam, 2006). When food is wrapped in green colors, this is a marketing strategy that is more likely to make it appear healthier or more eco-friendly than foods with alternate packaging designs. There's a

significant point on that as well. Such perceptions can even occur when there is no change in the product's objective products, that is, the product is the same, but the color is doing the talking.

A bit, colour can have a very strange effect on freshness perceptions. The bright, natural-looking colors are often seen as freshness, with darker or duller colors signaling good vibes about product quality. It is important for fruits, vegetables, drinks and organic products because freshness is essentially intertwined with quality assessments.

Also, sensory marketing research has demonstrated that colors in packaging could change up the taste expectation of the people. Just by looking at it, consumers may be expecting something sweeter, healthier, or more flavorful before any bite or sip occurs. Then those expectations become the real ones, perception and experience become a combined force – something like a loop – and yes, it's been

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emphasized as an effective interaction (Spence, 2015).

Thus, the food and beverage literature provides solid evidence of the impact that color has on quality perception expectations concerning naturalness, freshness, taste and health.

#### **4.4 Online Shopping Studies**

As ecommerce continues to expand, the importance of color becomes more apparent. With online shopping, consumers are not allowed to physically examine products and instead are more inclined to use visual information when choosing products to purchase. As a result, the colour palette of a website, the pictures of digital products and the design of the interface can really impact the perception of quality and trust.

Researches into online consumer behaviour indicate that aesthetics of a website can significantly influence consumer judgements. The visual design elements create first impressions and this

impression is often made in a matter of seconds after landing on a site. Colour is particularly significant as it can influence the impression of professionalism, credibility, and the credibility of the site (Hall & Hanna, 2004).

Blue is an interesting colour, and has become one of the more common colours used online, primarily in relation to trust and reliability. This is seen in many tech companies, financial institutions and ecommerce sites, where they employ blue schemes to create consumer confidence. Trust is closely tied to perceived quality, and color thus affects product evaluation in an indirect manner, through its effect on perceived credibility of the seller.

The screen color itself can affect users' experience, as well as their perceptions during the use of digital environments, as Gorn and his colleagues (2004) have demonstrated. Their findings suggest that the psychological response may be generated by visual elements without significantly affecting the content of the

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information. The same reactions can impact the user's experience when they are shopping online, and lead to a perception that the website is not polite, professional, or reliable.

Additional research on online retailing also suggests that consumers tend to bring their sentiments for or against the website into the buying process for products being sold online. A beautiful site with a clean color palette can enhance the perception of the quality of the products. Conversely, a poor visual environment can reduce consumer confidence in the retailer and the products. It can be very important in online sales; customers have no direct access to the intrinsic product characteristics, such as how it would feel or how it looks when seen in person.

Overall, the academic evidence reviewed here somewhat backs the theory that color effects consumer product quality perceptions in varying contexts. The research on packaging, luxury branding, food products, and online shopping pretty

consistently indicates that color leads evaluations with emotional appeal, influences expectations, fosters trust and infuses symbolic value. The strength of the effect may vary from one situation to another, but the review indicates that color as a real quality signal can have an impact, and individuals may use it when they are making judgments without much clarity.

## **5. INDUSTRY CASE STUDIES AND REAL-WORLD APPLICATIONS**

While academic research provides valuable insight into the impact of color on consumer perceptions, it can be beneficial to also observe real-world examples of brands to gain a deeper understanding of the application of color psychology. Many popular brands have intentionally incorporated specific colors into their branding, in order to send out a specific message, differentiate from competitors, and influence consumer perception of product quality. This doesn't necessarily

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determine the success of a business but when done regularly it can help to refine brand identity and cement brand associations in the minds of customers. The following case studies illustrate how organisations in different sectors use colour to create perceptions of quality, trust, luxury, innovation and value.

### **Case Study: Apple**

The best example of the use of color to "trick" the user into thinking it's a "premium" is at Apple. Since the late 1990s, the company has increasingly adopted a visual identity that is white, silver, and other neutral colors. That's where those shades fit in as part of Apple's brand image, which is all about simplicity, innovation, and an exceptionally good design.

One of the initial things people notice about Apple's brand is its simplicity. The graphics are generally subdued, fonts are clean and there is little use of background colour on these products. White is often

seen as a symbol of purity, clarity and simplicity. Silver, however is associated with modernity, precision and advanced tech (Labrecque et al., 2013). Together they have a "polished sophistication", but without using the darker shades that many classic luxury brands use.

If you have a look at Processing Fluency Theory, why this is the case seems to work quite well too. The design of Apple's look is simple and easy to comprehend, giving the user focus on the product rather than any additional graphics or ornamentation. Smoother "processing" conveys positive judgements and in turn promotes the notion of good product quality. Moreover, the brand has been used in various devices, retail spaces, and websites over time, making them increasingly familiar, and as a result, they seem more "trustworthy".

The color also helps Apple to be seen as an innovative company. Silver metallic finishes are usually linked with advanced technology, plus engineering precision. Because of that, people may start assuming



that Apple products are technologically superior, or at least more “mature” and sophisticated than competing products. And, importantly, these concepts tend to appear before consumers even look at the real technical requirements, or even listen to the actual performance of the product.

While brand loyalty and product functionality, as well as ecosystem integration, are part of Apple's reputation, its gradual color strategy does appear to be a big part for Apple. It can help to influence the perception of premium quality, innovation and a certain simplicity. Basically, Apple demonstrates how color can propel the larger brand narrative, and build additional quality-related connections as a more subtle but effective method.

### **Case study: Tiffany & Co.**

Not many companies show the power of color branding, like Tiffany & Co. does. One shade of blue, commonly referred to as “Tiffany Blue,” has become a

significant part of the luxury jeweler's brand. This tone is one of the most recognisable brand assets in the luxury industry over time.

The brand Tiffany & Co. has a blue that stands out from the black and gold palette fashioned by many luxury companies, and methodically distinguishes itself from the others. This colour has been used for packaging, advertising, in stores and marketing by the brand. Tiffany Blue is often synonymous with luxury, exclusivity, craftsmanship and quality to consumers.

Tiffany Blue isn't only pretty on the eyes, people seem to sense it immediately too. It's like the customers just click when I tell them the colour, before the brand name. That means that there is an unusually high correlation between colour and brand image. It's when the solid brand associations push recognition, trust and what people think the value is that brand equity is created (Keller 2013). In this way, Tiffany's choice of an eye catching, specific blue gives a full glimpse of the

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process, you can literally feel it where it touches.

Another subtle luxury signifier here is the colour. Not only the product, but the process of acquiring the product in that iconic blue box is now also becoming part of the overall consumption experience. The packaging says exclusivity and prestige, and it sets up expectations about the quality, even before consumers start to directly judge the jewelry. If you will, those expectations can increase perceived value from the get-go without anyone having a chance to “test” the product.

In the case of Tiffany, a lot of the other marketing messages have been removed, but even then, color can be a marketing tool that is connected to quality and luxury. With consistency over time, one colour could acquire strong symbolic value and this could impact consumer perceptions over the years and across generations.

### Case study Comparison: Color as a Quality signal

| Brand                                                                                         | Dominant Color                                                                                   | Quality Signal     |
|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|--------------------|
|  Apple       |  White/Silver | Premium Innovation |
| TIFFANY & CO. Tiffany                                                                         |  Blue         | Luxury             |
|  Whole Foods |  Green        | Natural Quality    |
|  Coca-Cola   |  Red          | Energy/Confidence  |

*Source: Author*

### Case Study: Whole Foods and Organic Food Brands

Another color-food relationship that is helpful is provided by the food industry. Like many organic and natural foods, Whole Foods Market utilizes the color green in their branding, and packaging. This is no coincidence. It mirrors bigger consumer tendencies and associations of green with health, nature, green living and eco-responsibility.

Some evidence suggests that, in this instance, consumers often interpret the green label as a signpost for the ingredients it contains and/or for the general healthier aspects of the product

(Aslam, 2006). This is especially the case for food markets where consumers will have limited ability to make any pre-purchase assessment of the quality and so will depend on what they can see before purchasing. In real life, visuals become somewhat “information substitutes.”

As for Whole Foods, it has been consistent as a retailer, seemingly emphasizing the natural and organic products. They are typically green in color, though, with their logo, “store” signage and the majority of their marketing material. This was really an echo of the same message, health oriented, eco-friendly consumption, health oriented, eco friendly consumption. Therefore, whenever you encounter these signs, your consumer may think that the food items you are buying here are fresher, healthier and quality in general, than what they would find in a regular grocery store.

The same strategies are used, more or less, by all organic food brands, worldwide. Sometimes, the green packaging is combined with images of greenery, leaves

on the farms and natural scenery. All of these elements together form a mini-story, an attempt to communicate a message of authenticity and purity. The color scheme can also have an impact on consumers' expectations of the quality of the product, even if they are not well informed about its contents or how it is produced.

Consider, as an example, the case of green branding within the organic food industry as a good illustration of how color-category match is important – and, yes, it really does. The brands that do meet the standards seem to receive a better quality and credibility rating because buyers typically believe that products with “green” visual cues are environmentally friendly. In that case, green doesn't simply serve as a nice background to be decorative, it serves as a hint, a sign that leads towards the product having those good, desirable attributes.

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### **Case Study: Coca-Cola and Pepsi**

Now Coca-Cola and Pepsi have been in competition for a long time, so you have got a pretty interesting indication of how different color identities can affect the customer's perception even if it's a fairly similar product category. Though both companies might be selling the same products, they may have used the other dominant color, thus producing different brand personalities in the mind of the customer.

Red has been Coca-Cola's primary colour for a rather long time now. As mentioned earlier in the paper, the color red has been associated with excitement, energy, passion and stimulation (Elliot & Maier, 2014). This connection effortlessly aligns with Coke's advertising campaigns, which have been all about happiness, celebration and emotional experiences. In practice, the company's reds are dynamic and full of life, and therefore reflect the type of image that appeals to consumers around the world.

Pepsi, however, has always employed blue as a sort of brand color. Trust, reliability, calmness and competence are among the qualities that are often associated with the colour blue. Each of these ideas adds a certain sweetness to the brand personality, although less than Coca-Cola's. Blue has frequently been the colour to convey modernity, youthful energy and novelty, a role which Pepsi has had to fulfil many times.

Of note: there is no strict definition of “higher objective quality” for either colour. Instead, each hue moves the consumer's definition of the brand and the products that it represents. The red colours for red brand mark Coca-Cola can make the customer think of excitement and emotional involvement; the blue colours for blue brand mark Pepsi can make the customer think of stability and a more modern style. It's not just that those links may influence consumer behavior and consumer purchases, but that they can also

do so if the actual difference in quality is minimal, or even impossible to assess.

This is sort of a Coke and Pepsi example of using color as a tool strategically and dividing. If the two firms maintain their own colour schemes, then they develop different brand personalities which help consumers to understand what they are purchasing. Overall, it implies that color does not just influence the quality judgment per se, it also expands the brand associations which subsequently, indirectly, impact the quality judgment.

When you add these case studies up, it's easy to see that there are color psychology applications that span the range of industries. From Apple's strategy of introducing a new "white and silver" color in its iPhone line to signal innovation and premiumness, to Whole Foods' strategy of employing a particular shade of green as a reminder of health and naturalness, to even Coca-Cola and Pepsi's distinct color identities to help maintain their brand personalities. Each of these businesses

could be selling a different product in the same or different market, but they are all examples of a much greater concept: color is a powerful visual cue that can have an impact on the consumer's perception of the quality of the product. In the end, these strategies will work well if they're consistent and work with the category and if the color can continue to serve the brand's desired positioning.

## **6. MODERATING FACTORS AND LIMITATIONS**

As indicated in previous sections, the evidence indicates that colour can have a significant impact on the consumer's perception of product quality. Color is a big visual cue and works through emotional connections, mental shortcuts, category expectations and brand identity to help consumers make decisions when the situation is a bit confusing. Yet, claiming that color is all that matters in determining quality perceptions is erroneous. The

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impact of colour does not have a consistent effect and can change in each consumer, product and context. Instead, they are influenced, or even “moderated,” by several factors that influence the perception of colors and the values attributed to them in decision making. Understanding these boundaries appears to be crucial to correctly respond to the primary question of the research: What is the degree of impact that color psychology has on perceptions of product quality?

### **6.1 Cultural Differences**

The one of the most important color psychology restriction is that colors meanings are somewhat culturally dependent. Many associations people make with colors are based on cultural traditions, social norms and historical events, so while some may appear very common, many are not. So, there is a possibility that one color that makes good quality impressions in one cultural context may have completely opposite interpretations elsewhere.

A frequent case in point is white. In many Western societies, white is associated with purity, cleanliness, simplicity and even safety. Which is why white is frequently chosen for medical applications, electronic devices and personal hygiene products. People usually take white packaging as a hint of hygiene, reliability, and high quality. Nevertheless, white has been traditionally associated with the concept of sadness and death in some of the East Asian cultures such as China and parts of Korea (Aslam, 2006). As a result, the emotional responses that can be elicited by white may vary, or even be completely different, across different cultural settings.

Such trends can be observed for other colors as well. In many Western countries, red is associated with danger or caution, while in Chinese culture, it's connected to prosperity, good luck, and celebration. While green may represent environmental sustainability in many western markets, its interpretation may vary based on the local culture and its associated past experiences.

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All in all these cases demonstrate that color significance is not predetermined, it is more like being influenced by the social and cultural context.

In a sort of indirect way, research on cross cultural color preferences confirms that. Similarities and differences were detected in the color meanings across the various countries by Madden et al. (2000), and it appears that, yes, there are color meanings that are shared across a range of cultures, but there is cultural variation in how the basic idea is negotiated. Therefore, marketers must not take it for granted that a color strategy that is effective in one market is the same in another, as in fact it isn't. That is, there remains a cultural context that can be a moderating force for color effect.

## **6.2 Gender and Demographic Variation**

Demographic factors like gender, age, education and experience also influence how consumers will use color. While some general patterns can be found, there are

also significant differences in attitudes and responses of different consumer groups towards the colors.

Conducted research on colour preferences has shown that there may be gender differences in colour preferences and evaluations. For instance, Elliot and Maier (2014) report that women tend to be more adept at noticing color differences, and may have more skills in discriminating for shades and hues. It isn't just a biological preference that affects their love of a particular colour; sometimes it may be a cultural and social preference as well.

Age can also affect colour perception, or be a factor in colour perception. Bright and brightly colored colors are more appealing for younger people in any entertainment, fashion or tech item. For older folks, on the other hand, they are more likely to choose more subdued and traditional shades, frequently linked with dependability and familiarity. However, there are numerous other factors including education, previous buying experience,

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lifestyle etc, that can have a subtle affect on how a visual signal is interpreted.

With these demographics changes, one has to think of color psychology as not a “one size fits all”. Things that appeal to one group of consumers may not appeal to another. Therefore, businesses have to do proper research and consider the audience profile to create color-based branding, and select the packaging strategies.

### **6.3 Product Category Effects**

Another restriction is due to the product classification. As in all cases, the meaning of a colour may differ from one case to another and the same colour can give rise to varying quality evaluations if the consumer is evaluating various qualities.

For example, black is a good one. Black can be considered as a luxury element for luxury products as it gives a sense of refinement, luxury and exclusiveness. For luxury watches, luxury cars, cosmetics and fashion products, a black package typically expresses the luxury status. Such

correlations have become ingrained in the minds of people, hence black is a good quality signal in luxury segments.

The same color can have a different mood in different settings, though. For example, a black milk carton could be a bit unusual, or perhaps even incorrect for some consumers; milk products are something that often brings into mind freshness, purity and cleanliness. White or light packaging would be a better option, on the other hand. So it's not just about the color of the product, but whether black is a quality hint or not.

This sort of thing is an obvious indicator of colour – category congruence, mentioned previously in the paper. People have expectations for what colours are suitable for a particular product category, and those expectations affect their verdict on that product (Bottomley & Doyle, 2006). If a color does correspond to the normal pattern, then people tend to read higher quality. However, the color that goes against the rules may end up lowering



quality impressions. In short, there is no universal meaning of color; it depends on many factors, particularly the appropriateness of the setting.

#### **6.4 Brand Reputation Overrides Color**

While color may influence first impressions, it is somewhat less influential when consumers have a solid knowledge base of a brand. When a brand has a good reputation, there is another, not so obvious, information channel that the customer enters, and that can be more important than visual.

This kind of trust, familiarity and credibility is built up over time, which is important for well established brands. If consumers already think a firm's products are top notch, they may not be as concerned about the color of the packaging as much as they would if they were not sure. As with Apple, customers are known for their loyalty, love of innovation and great product, and those things are supported by years of product experience

and customer feedback. While it helps, the majority of folks don't rely on color as their sole method of determining Apple products.

Luxury brands such as Tiffany & Co. appear to extract perceived value beyond the surface, too. Their heritage, craftsmanship and overall reputation do much of the work. The iconic blue package keeps that notion alive, and that brand power isn't limited to color either. So often, consumers have preconceived notions about the name and its prestige and quality before they see the visual cues.

There's also work in brand equity that supports this. According to it, strong brands reduce consumer uncertainty as they provide consistent cues regarding the product performance that should be felt by the consumer (Keller, 2013). In reality, the brand will be more important to buyers than the color, particularly for products they're familiar with. This is somewhat restrictive, as other information sources are often lacking, delayed, or not present,

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thereby making the use of color most effective.

### **6.5 Risk of Overestimating Color Effects**

But, perhaps the most important constraint is that people are prone to overestimate the role of colour in consumer decision making. While color may influence perceptions, quality judgments are truly multi-dimensional and based on many factors that are not directly related to what's seen.

Prices can frequently be one of the best “signals” of quality in many markets. When objective quality evidence is difficult to understand and/or substantiate, shoppers frequently equate price with product quality, particularly when the price is high (Zeithaml, 1988). Another important one is online reviews and ratings, which allow individuals to get an idea of what they received from an individual before they make their purchase. Brand image, promotional messaging, the country of origin, and even

experience with similar products, can all affect the interpretation of quality.

Most importantly, real product performance finally steers the longer term evaluation. On first sight a product could seem like a quality product, only to be quickly forgotten by the user because of the packaging color. Meanwhile, something that's less visually appealing can actually gain a good reputation because it has been producing good results time and time again.

The relationship between color and quality perception should therefore be seen as more probabilistic, not totally deterministic. Color can nudge expectations, sets up first impressions early, and it can sway buy decisions, especially when consumers don't have complete information. However, it is typically not a one-way process. Instead, color combines with a variety of other signals, and these in combination form the perception of quality.

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In combination, these “in-between” factors suggest that color psychology does indeed influence consumers' perception of product quality, albeit somewhat less than other factors. How consumers interpret colors, and whether they even do, varies by culture, by demographic, by product category, by brand reputation and more. Thus, colour should be considered as only one of the whole system of cues which the consumer uses in assessing products and brands.

## 7. DISCUSSION

This study was done to examine to what extent color psychology works to either mislead or manipulate consumer perceptions of product quality. During this paper, I studied a lot of research on color and it's quite obvious that colour is not just a decorative feature on a product or a brand. More like color can be a good visual cue and it is used to make decisions on topics such as quality, reliability,

luxury, security and value. In all the differing theories, the empirical studies and the examples from industry, it is the same pattern that keeps appearing: color leads consumer perception, influences emotional response and mental judgment in the first place before they really taste, feel or touch the product. However, the concept of automatic is not proven to be true in all instances. The impact of colour is determined by the context of the product, cultural references, the nature of the product but also if there are additional signals in the context which are also “quality related”.

A generalized rule in the literature is that colour is a form of an immediate visual cue in consumer decision-making. In many cases, there are times when the consumer has to decide what to buy without really knowing exactly what they may be getting, particularly if they are buying the goods online – where they can't inspect the goods themselves before buying. Visual cues are even more critical in such situations. The

basic premise of Signaling Theory is that when consumers cannot have all the information they need, they use what little information is available to guess at the quality of the product they are considering buying (Spence, 1973). One of those signals is then the color; color is a symbolic sign that consumers relate to certain attributes of the product.

The discussion of theory's we talked about before sort of explain why these effects occur. Color psychology research shows that colors trigger emotional associations that grow out of cultural learning and repeated exposure. Colors are associated with trust with blue, competence with black, naturalness or health with green, and excitement or energy with red. These feelings then influence expectations of a product or brand before objective product characteristics are judged. Consumers may deem higher quality products, simply because the presentation feels like the psychological associations.

This is also corroborated by the results of the Cue Utilization Theory. When there are missing or difficult-to-assess intrinsic cues, consumers often rely on extrinsic cues such as packaging, color, and brand (Olson & Jacoby, 1972). As it is not always possible for consumers to assess product quality directly before buying, they rely on visual information to develop expectations. Color in particular is a very practical shortcut, reducing uncertainty and simplifying the process of making decisions. The evidence that was reviewed in Section 4 supports the fact that consumers are able to reliably check products based on color cues across various industry categories, and not just one.

The best evidence for the effect of colour on people is seen in research on packaging and product presentation. Orth and Malkewitz (2008) discovered that visual package design can actually alter brand perceptions, in particular, the perception of the brand being “sophisticated, competent,

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and sincere”. Similarly, findings with respect to color congruence highlight a similarly consistent finding: that consumers like products more when the colors of the packaging match the colors that they are accustomed to seeing for that particular product category (Bottomley & Doyle, 2006). Combined, these findings suggest that color influences quality judgments not only due to its emotional appeal but also its cueing of the sorting function—consumers are able to determine whether something looks appropriate and credible for its category.

The weight of color becomes more evident once it becomes a topic of food and drink. Research indicates that colour is a cue for freshness, health, naturalness and taste expectations (Spence, 2015). Consumers, for instance, associate green packaging with more natural ingredients and that good packaging is better for the environment – before it even reaches their hands. Reactions occur early, before interacting with the product, which means

that the visual cue has a great amount of work to accomplish. However, there is also a similar feeling in luxury spaces with the black, gold, and dark colors being connected to luxury or high-end. Industries, more or less simultaneously, introduce colour as a language, to indicate product properties and position in the market by colour.

Not only do the findings support the studies being done on color psychology, but the industry case studies also are illustrative examples of the way color psychology is being utilized by businesses.

The combination of white and silver conveys a sense of innovation, simplicity, and premium quality, which is crucial for Apple. Thus, the brand's signature blue has basically become a symbol of luxury and exclusivity in the Tiffany & Co. brick-and-mortar boutique and, in this instance, it has gained a symbolic value over time. Green is used by the likes of Whole Foods and other brands of organic food as a representation of health and environmental

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sustainability, while red is used by Coca-Cola and blue by Pepsi to evoke a distinctive brand personality. In general, these examples show that there is a large potential of colour use in brand issues, and colour can impact the consumers perception in a number of market sectors.

However, despite this proof, the outcomes also highlight the important shortcomings, which suggests that colour psychology is not a panacea for quality perceptions. Culture is one of the major factors that moderates. As mentioned above, colors may have multiple interpretations in various cultures. In the west, white symbolizes purity and cleanliness while in some Asian cultures, white is the color of mourning. Likewise, the meaning of red may vary between negative and positive, depending on the context. This means that this sort of colour effect is not universal, or the same in all places and contexts, but rather is influenced by social and cultural learning. Therefore, it is important for marketers to take cultural context into

account when planning color strategies and not presume that the same reactions will occur in the various geographical areas.

Another important limitation, deals with demographic variation. Color preferences could differ depending on gender, age, experience and lifestyle. Not all consumers can read colors and preferences differ greatly across consumer groups. This type of variation can make color impressions more challenging to predict, and can definitely highlight the importance of understanding your audience when producing your branding and packaging.

Likewise, the success of colour cues in this context seems to be extremely consistent cross product types. It was suggested that there should be a correlation between the colors used and the category of products they belong to, known as color–category congruence, whereby consumers believe that colors are being applied to a product type. Black might be a sign of luxury items, green of quality items in the organic food market. However, the same tones can

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prove less than effective—or even backfire—when applied to the wrong type of product category. “power” of colour is not only the colour itself but also the context in which the colour is used.

Perhaps the most important lesson to be learned from this study is that color seldom works in isolation. Colour might make the initial impression, quality might be among the checks, but there are many other signals that shoppers pay attention to. Though, price remains one of the best bet predictors for quality as many consumers equate quality with higher price (Zeithaml, 1988). For instance, when buying a product, its experience, product specifications, warranties and reputation are more significant than the color of the product. For strong brands, consumers may not be so reliant on colour cues, as they tend to have some notions about quality and reliability without them.

This is relevant to the answer to the research question. The evidence here suggests that colour psychology can

influence the quality perception of a product especially in the early phases of consumers' initial quality assessment, when there is limited information and visual cues dominate the entire quality assessment. But that influence is to be seen as complementary and not as the ultimate. Colour does not substitute for all the other cues of quality, but can be used to bias expectations, draw attention to movement and suggest the symbolic meaning. It operates instead of price, reputation, product performance and even the user experience, in a more holistic quality measurement framework.

Overall, the results suggest that color psychology may be a relevant, but context-dependent tool for influencing the judgment of product quality. The effects appear to be strongest when consumers are very visual and when the colors they select are those they expect for the category, and when their signals of quality are fairly limited. The other way around, the influence is reduced when the consumers

already have a lot of knowledge of the product, or when the stronger signals (the brand, its performance, whatever) can be more easily detected. Color, yes, but it is just one piece of a complicated puzzle that together is perceived as product quality.

## **8. CONCLUSION**

The purpose of this paper was to investigate the extent to which the influence of color psychology can influence consumers' perception of product quality. The findings, which are based on a combination of theory, empirical research and some form of a real industry scenario, indicate that indeed, colour has a significant impact on product and brand perception. While many consumers may state they base their purchase decisions on objective features, such as performance, durability, and the product's function, the evidence suggests that many times “quality” is interpreted before having any real hands-on experience. At the point

when there is little information or little time, consumers resort to what they see and make those decisions quickly in lots of buying situations. Of these visual cues, color seems to have a significant impact on initial impressions, and initial quality ratings.

Through this study it was found that the literature showed that color does have an impact on the consumer perception but in not such a neat way. Rather, it appears that it functions through a number of linking channels, more or less. First, colors incite emotional responses, and the emotional responses drive the understanding, interpretation, and evaluation of a product. For instance, colors such as blue, black, green and red tend to evoke feelings of trust, sophistication, naturalness, excitement and reliability. Many of the emotional cues actually bias product quality judgments, too; and, they inadvertently set expectations for what the product should feel like and how it should function.

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Second, color can act as a sort of mental shortcut, so people can make fast evaluations when they do n't have enough information at hand. Consumers tend to rely on visual cues when guessing the level of quality rather than making long detailed checks of product attributes. Third, color is a quality signal by Signaling Theory and Cue Utilization Theory, which allows consumers to develop expectations about the product without actually purchasing it. Last but not least, when the brand consistently employs color, it aids brand identity, familiarity and trust, which will have an indirect effect on perceived quality.

Academic research reviewed provided fairly good evidence that there is a link between color and quality judgments. The researchers find that color has a real impact on evaluations, such as competence, sophistication, and even the “overall appeal” feeling, when they examine the design of the packaging. In the luxury branding research, the darker

the color, the higher the prestige and premium quality – the black color and the gold color are basically directly linked to higher quality and status. Color also influences perceptions of freshness, health and flavor in food and beverage markets – which is a big deal when making daily choices. The same applies to online shopping environments: website colours and how products are presented will influence consumer trust and increase the professional appearance of a website. All of these results suggest, however, that color may be a ubiquitous element, in many purchasing contexts.

The industry case studies took it one step further and basically demonstrated how organisations actually plan to use colour in a systematic manner to impact on those perceptions. The use of white and silver, a palette that can be seen as Apple used with its own products, presenting the image of innovation and quality. Meanwhile, Tiffany & Co. take on its blue signature, standing for luxury, exclusivity. Green is

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used by whole foods and other organic brands to convey health, sustainability and naturalness – whereas Coca-Cola and Pepsi use contrasting colour identities to make their brand personalities stand out. Overall, it demonstrates that colour is not merely a decorative element; it is, indeed, a strategic tool that can enhance brand positioning and sway perceptions of quality.

Furthermore, the study identified some of the main challenges and problems, in a certain way which reduced the importance of the role of color psychology. Personality differences are brought out, people don't read colors equally. Therefore what may be seen as “quality” in one instance could be interpreted differently in another. Furthermore, there are differences in consumer reaction depending on consumer characteristics (age, gender, experience, etc.). Another product type question: Color is not all-important for all products. A shade that implies quality a good deal in one context may not be that strong or even

backfire in another. Perhaps most importantly, most of the time color is not used alone – consumers are also using many other cues, including price, brand reputation, reviews, advertising and prior experience.

The evidence indicates that colour may be an important factor in evaluating the quality of products especially when the consumers have no prior knowledge of the product and only rely on the visual information. Color evokes emotional reactions, builds trust, communicates symbolically and serves as a quality indicator. But this influence can not be forever. It depends on the cultural context, the type of product, the characteristics of the consumer and if other indicators have already been established and are the key driver, such as price or the reputation of the product. While color psychology is a great marketing and branding tool, it should be used with other factors of influence working together, and not as a

single factor, to determine perception of product quality.

To be honest, the results of this paper essentially underline the fact that visual communication is a key element in today's consumer markets. Digital commerce is still in its growth stages, and consumers are looking to sight over sight to make their purchasing decisions, thus the innovative use of colour will continue to play a vital role, gently influencing consumer decision making and opinion. Maybe future research can further explore the role of new technologies, evolving cultural values and consumer preferences in this, and how all of this affects the quality perception of the product.

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